

Controlled chemical worlds reveal limits of generalization in autonomous scientific research

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Abstract

Autonomous scientific research requires deciding which experimental relationships generalize to new conditions. This ability is difficult to evaluate when mechanisms are unknown and agents select their own evidence. Here we introduce ChemWorld, a programmable framework separating chemical process laws, experimental observations and supplied information. Across 240 autonomous campaigns in six system families, operational achievement and predictive knowledge diverge. Electrochemical optimization improves recommendation performance in 26 of 30 matched comparisons, but prediction in only 14. Larger budgets improve electrochemical and partitioning predictions, while crystallization errors remain response dependent. In aqueous equilibrium, aligned information improves predictions on nine questions but worsens them on the three most dilute conditions in every tested world. Conversely, crystallization predictions unnecessarily depart from stable observed purity. These complementary failures identify the applicability of empirical relationships as a central evaluation problem for autonomous science, beyond successful operation or plausible mechanistic explanation. Autonomous laboratories and language-model agents increasingly connect scientific planning to executable experiments¹⁻³. Their promise extends beyond finding a productive recipe: an autonomous researcher should learn relationships that remain useful when conditions change. Yet successful operation and scientific understanding have different requirements. One high-performing condition may suffice for an optimization task. Predicting an unfamiliar intervention requires distinguishing what the observations establish, which alternatives remain compatible with them and where an inferred relationship may cease to apply.

This distinction is difficult to investigate in physical laboratories. Mechanisms are incompletely known, identical starting states are hard to reproduce, and an agent’s prior information changes which evidence it acquires. A failed prediction may reflect missing experiments, unresolved identifiability or an inference that departs from available observations. Conversely, a correct prediction may follow a simple regularity without identifying its mechanism. Neither task scores nor the apparent sophistication of an explanation resolves these possibilities.

Computational environments offer complementary control. Optimization benchmarks provide repeatable objectives^{4,5}, and interactive chemistry and discovery environments support sequences of scientific actions⁶⁻⁸. The opportunity is to combine composable physical processes, persistent experimental state and controlled information with assessment of what an autonomous research process produces. Such an instrument must preserve freedom to select experiments and formulate explanations, while making the underlying laws and observations available to the evaluator.

We introduce ChemWorld and use it to examine this relationship across six chemical system families. The platform spans nine task families; the agent evidence reported here covers electrochemical conversion, phase partitioning, reaction and thermal processing, aqueous equilibrium, crystallization and purification. We follow a single agent configuration through experimental acquisition, a free-form mechanistic report, prediction at withheld conditions and retrospective questioning. The resulting evidence shows both successful generalization and distinct failures to preserve or revise empirical relationships. These are observations

within controlled models, intended to motivate and localize further study of autonomous scientific inference.

Results

A programmable instrument separates laws, observations and information

ChemWorld constructs executable worlds from compatible process and observation components. Each world specifies persistent material states, legal operations, instruments, finite resources and private parameters. Operations can change composition, phase allocation, thermal history and remaining material; measurements expose declared quantities rather than the full simulator state. The evaluator can vary private laws independently of public descriptions, or change supplied information while keeping a world fixed (Fig. 1).

The frozen platform qualification passed 64 task-world units, 1,786 boundary and categorical recipes, 192 invalid-action probes and 52 generated compositions. The generated cases comprise 18 previously absent topologies, eight new task-world identities using a registered topology and 26 additional coverage cases. Component and interface checks passed 32 module probes, seven interface paths and seven invalid compositions. These finite-domain checks establish internal model and execution consistency; they do not establish agreement with arbitrary wet-laboratory chemistry. Exact environment replay reconstructs executed actions and resource consequences, including rejection and rollback.

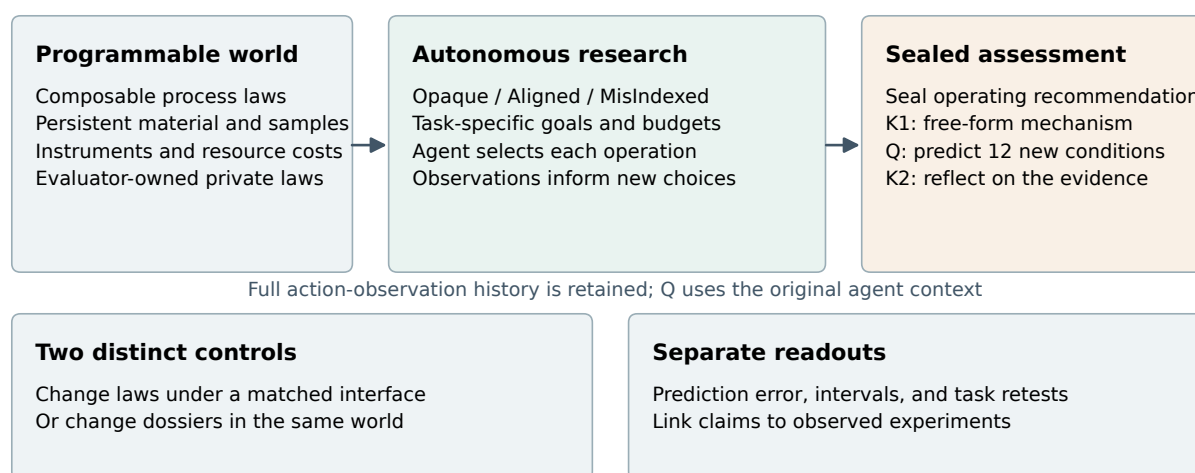


Figure 1: ChemWorld separates the physical world, public evidence and assessment. Information treatments share a physical world. Agents choose experiments and express mechanisms freely. The operating recommendation is sealed before the mechanism report and evaluated in a separate retest. Predictions retain the full experimental context, and retrospective questions follow the sealed predictions.

The research programme contains 240 campaigns, each belonging to a five-world, three-arm study block. Opaque supplies usable public contracts and anonymous identifiers. Aligned adds the tested nominal information; MisIndexed supplies its fixed mismatch or reassignment. Information can concern entity associations, effective parameters or structural relationships. Aligned information is partial, and Opaque does not remove general chemical knowledge. These distinctions avoid equating any dossier with complete knowledge of the simulator.

The agent freely allocates 12 or 24 batches and a matching allowance of additional measurements. Objectives are tailored to the system: discovery and optimization are separate commissions in electrochemistry and reaction processing; partitioning and equilibrium emphasize characterization; crystallization and purification seek recovery subject to product-quality constraints. All 720 report, prediction and retrospective

stages completed. Two campaigns fell short of their final-assay budgets, leaving 238 conforming chains and 3,597 of 3,600 planned source assays. Both remain in the results.

Operational achievement and predictive knowledge separate

Electrochemical optimization improves the independently retested recommendation in 26 of 30 comparisons matched to discovery by world, information arm and budget. Score-prediction error improves in only 14. Thirteen comparisons combine a better recommendation with worse prediction, and these discordances occur in all five worlds (Fig. 2). Restricting to pairs without infrastructure-driven source restarts retains the separation: 15 of 17 improve recommendation performance, eight improve prediction and seven are discordant.

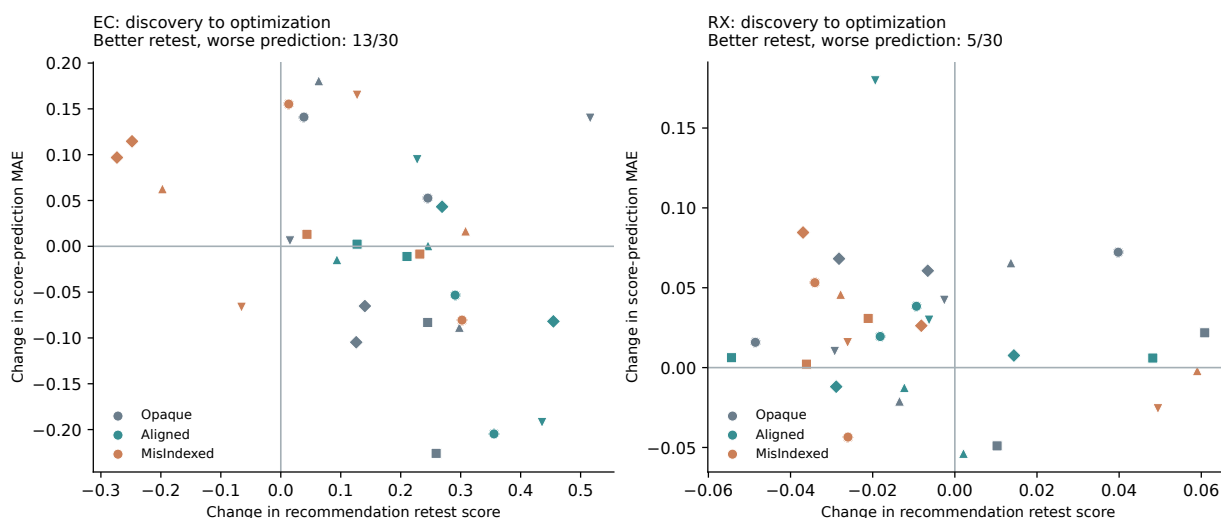


Figure 2: Task achievement and prediction are separate outcomes. Each point pairs optimization with discovery under the same world, information arm, budget and prior locus. Electrochemical and reaction-processing panels use their own public score scales. The goal intervention changes the research commission; the comparison does not isolate experiment selection from subsequent inference.

The effect is not universal. In reaction processing, optimization improves retests in nine of 30 comparisons, score prediction in eight and the original six-response macro error in six. Thus the optimization commission itself does not guarantee better delivery. Prediction and decision losses need not align⁹, and these experiments show why both must be measured in an autonomous research workflow.

Purification gives a related distinction between recovery and acceptable delivery. Aligned recommendations meet the purity threshold in three of five worlds, compared with none for Opaque or MisIndexed. Yet mean recovered fractions are 0.04455, 0.12373 and 0.32105 for Aligned, Opaque and MisIndexed, respectively. Recovering more material therefore does not imply satisfying the commission. A single retest near the purity threshold is also not a precise estimate of a procedure’s reliability. These system-specific outcomes cannot be represented faithfully by a universal yield score.

Additional experiments help selectively

Increasing the research envelope from 12 to 24 batches substantially improves several predictive outcomes (Fig. 3). Electrochemical score mean absolute error (MAE) decreases from 0.17378 to 0.11222 under discovery and from 0.17811 to 0.10818 under optimization. Eleven and twelve of the respective fifteen matched campaign pairs improve. All six electrochemical responses improve in aggregate, and their mean directions persist when each world is omitted in turn.

Phase-partitioning organic-fraction MAE decreases from 0.08291 to 0.02379, with improvement in 13 of 15 pairs. Correct process-choice answers increase from 25 of 30 to 30 of 30. Nominal 90% interval coverage rises from 78.3% to 98.9% while mean width narrows from 0.25122 to 0.18553. This combination supports better prediction, although coverage above the nominal level is not perfect calibration. In a selected aligned campaign, the larger budget covers all sixteen material pairings and then allocates eight experiments to process variation; the smaller campaign covers ten pairings. This is a plausible route to improvement, not a cohort-wide causal explanation. Two aligned world pairs nevertheless worsen.

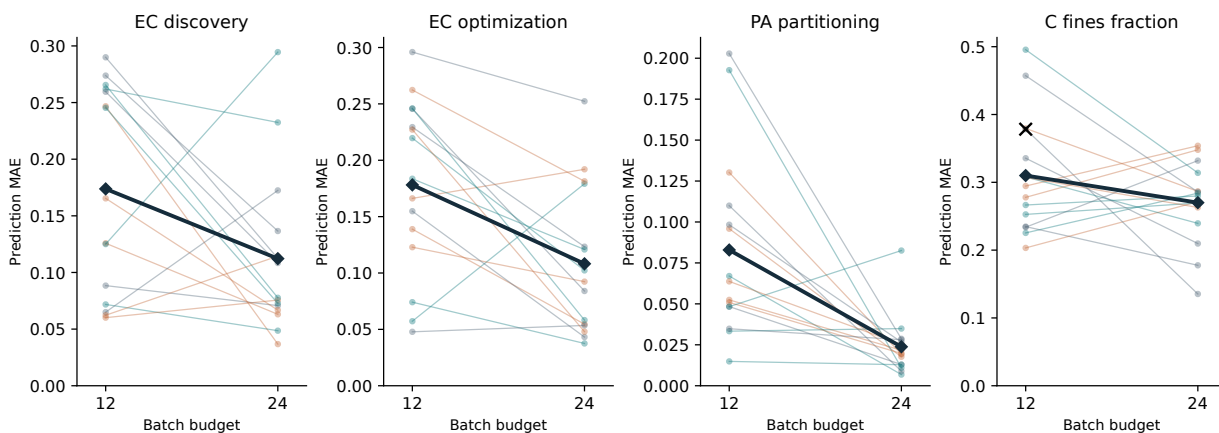


Figure 3: Budget effects depend on the response and commission. Matched 12- and 24-batch campaigns are separate research sessions. Additional batches accompany larger operational and computational allowances, so the contrast measures a research resource envelope. Readouts have different scales; response-specific values and interval summaries are reported in the supplementary tables.

Crystallization behaves differently. Net-recovery MAE decreases from 0.10433 to 0.07937, but quality-eligible recommendations remain 12 of 15 at each budget and mean retested recovery decreases from 0.42054 to 0.39939. Fines-fraction interval coverage improves from 35.6% to 43.9%, still below the nominal 80% in every campaign. Larger resources help some forecasts without resolving all response errors or improving delivery. Because the larger condition is a fresh session with additional computation, these contrasts do not identify the isolated value of twelve extra observations.

Information value changes across intervention regimes

Prior information has demonstrable benefits. In the reaction parameter study, aligned information lowers discovery macro MAE from 0.10186 to 0.05701 and optimization error from 0.13893 to 0.08384, beating Opaque in four of five worlds under each commission. Across other blocks, however, there is no common ordering of the three arms. Close aggregate means in the equilibrium entity and structural studies are not evidence of equivalence.

The equilibrium parameter study gives a sharper pattern. Overall macro MAE is 0.01214 for Opaque, 0.04314 for Aligned and 0.03928 for MisIndexed. Opaque outperforms both dossier conditions in all five worlds. A post hoc grouping by public recipe concentration locates the reversal (Fig. 4). On the nine questions other than the three most dilute, Aligned improves MAE from 0.01008 to 0.00465. On the three most dilute, its MAE instead rises from 0.01833 to 0.15862. Both directions occur in every world. The remaining nine questions include boundary conditions and should not all be called interpolation.

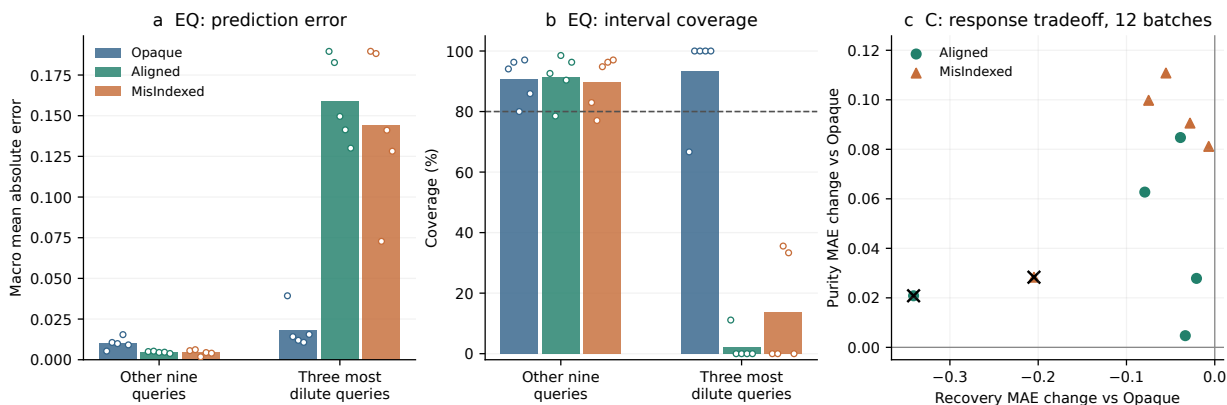


Figure 4: Information effects depend on intervention regime and response. Left: equilibrium parameter-study macro MAE on the three most dilute public recipes and the other nine. Centre: interval coverage on the same groups; the dashed line marks nominal 80% coverage. Right: crystallization, 12 batches, paired differences from Opaque in recovery and purity MAE. Every displayed dossier pair improves recovery but worsens purity; crosses mark pairs containing the eleven-assay source. Equilibrium groups and the joint crystallization comparison are exploratory retained-data analyses, with five reused worlds per study.

Aligned intervals also fail selectively: coverage is 91.3% on the other nine questions but 2.2% on the dilute three, versus 90.7% and 93.3% for Opaque. The dilute recipes have nominal concentrations near 10^{-4} to 10^{-5} M; all fifteen source campaigns remain above 0.018 M. Observed dissociation therefore occupies a plateau near 0.06–0.09, while the dilute tests require leaving that regime. Broader input coverage alone does not solve this problem. In one world, Aligned explores a ninety-fold concentration range and Opaque a fourfold range, yet both remain on the observed plateau. At the most dilute test, the reference dissociation fraction is 0.65463: Opaque predicts 0.629 with interval [0.45, 0.80], whereas Aligned predicts 0.0804 with [0.053, 0.108]. This selected example illustrates the aggregate pattern without establishing that the dossier causally produced anchoring.

Stable observations can also be abandoned unnecessarily

Crystallization provides the complementary problem. For every campaign, we construct a mean predictor from its own public final assays and a nearest-neighbour predictor using public recipe features. These references acquire no additional experiments. Agent recovery forecasts outperform the observation mean in 26 of 30 campaigns, but purity forecasts lose in all thirty; size forecasts lose in 26 (Fig. 5). This selectivity argues against interpreting the result as a general inability to use data.

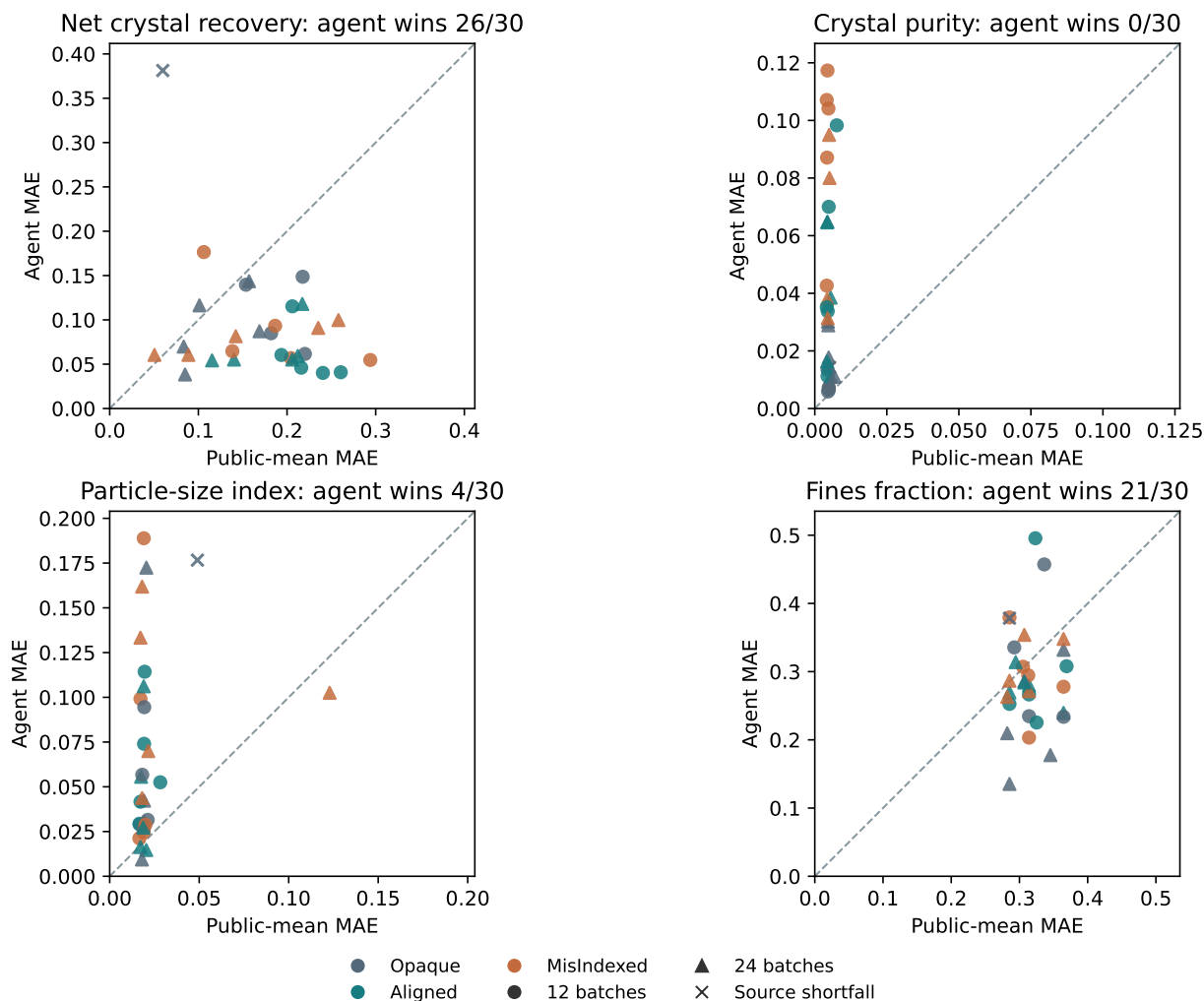


Figure 5: Prediction compared with each campaign’s own observed data in crystallization. Points above the diagonal favour the public-observation mean. Agents improve net-recovery prediction in 26 of 30 campaigns but lose on purity in all thirty. Symbols distinguish budgets, information arms and the nonconforming source. Purity varies little across the withheld queries, making a constant forecast a strong reference; success on that response alone would not demonstrate mechanism discovery.

Source and withheld purity average approximately 0.985, whereas agent predictions average 0.93544 at 12 batches and 0.94893 at 24. Of 360 query forecasts, 335 are too low. The mean within-world withheld-purity span is only 0.02442, explaining the strength of the simple reference. In a retrospectively selected misindexed campaign, the mechanism report explicitly acknowledges that only one solvent was tested. Nevertheless, predicted purity averages 0.86833 against a reference mean of 0.98566, and all twelve nominal 80% intervals miss. A verbal statement of uncertainty does not by itself ensure calibrated quantitative prediction.

The same information condition can help one response and hurt another. At 12 batches, both Aligned and MisIndexed improve recovery MAE while worsening purity MAE relative to Opaque in every crystallization world. The direction holds in all four both-conforming pairs for each comparison. It weakens at 24 batches, so it is not a budget-invariant law. In equilibrium, some predictions preserve an observed plateau beyond its applicability; in crystallization, predictions depart from a stable observed response without improving accuracy. Together, these findings identify the decision to preserve or revise a relationship as a useful object of study.

Discussion

ChemWorld connects the controllability of computational models with the open sequence of decisions made during autonomous research. Its contribution is both an experimental instrument and evidence that operating success, predictive generalization and warranted uncertainty can diverge. Positive partitioning and reaction results are essential to this interpretation: the observed difficulty is selective, rather than uniform failure across scientific tasks.

The complementary equilibrium and crystallization patterns suggest a question beyond whether an agent can fit observations: can it identify which relationships should change under a new intervention? Copying an observed mean works unusually well for crystallization purity but fails badly on dilute equilibrium tests. Neither a universal preference for simpler empirical prediction nor a universal preference for mechanistic extrapolation follows. The useful capability is to connect an extrapolation to the regime and evidence that justify it.

These results do not yet isolate the causes of that capability’s failure. Information arms alter the evidence agents acquire as well as their later interpretations. Predictions retain the full source context, so the study does not measure information loss caused by compressing research into a report. A natural-language account may omit a useful relationship that is reconstructed during prediction, or acknowledge an uncertainty that the forecast fails to represent. Prospective evidence-matched interventions and independent report recipients would be needed to distinguish these mechanisms. Retrospective interviews alone cannot establish them.

The empirical scope is also bounded. One agent configuration and five world instances per study cannot establish a universal limitation of language models. Related observations, responses and arms are not independent replications. Query distributions determine which regularities are useful; the dilute subgroup was identified after inspecting outcomes. The simulation’s finite validity domain and absence of wet-laboratory validation limit transfer to physical discovery. Protocol differences across system blocks further preclude a pooled accuracy ranking. Within these limits, explicit controls and complete outcome records make the results testable and expose where stronger experiments should intervene: between acquired evidence, claimed applicability and predictions of new conditions.

Methods

World construction and execution

A world is specified by public experimental contracts and private process parameters. Process components declare state, domains, dependencies and compatible interfaces. Operation and observation models compile these declarations into a laboratory interface. The participant can act through that interface but cannot inspect the simulator, private parameters, evaluator records or repository. Instruments may consume samples and operations consume declared material, time and action allowances. Candidate transitions are validated before commitment; rejection preserves committed physical state while retaining the declared attempt consequences. Replay concerns the environment and recorded operations, not regeneration of language-model responses.

Qualification distinguishes internal physical and accounting consistency from successful research. It checks executable domains, resources, information separation, valid query references and exact replay. Invalid actions and incompatible compositions have explicit expected rejection or rollback behaviour. Subsequent study adapters have their own checks; the frozen platform qualification is not an automatic certificate for later execution code. The reported autonomous campaigns retain their development-evidence status and are not promoted to formal validation by this analysis.

Study units and objectives

The eight blocks comprise electrochemistry with entity priors (60 campaigns), partitioning with entity priors (30), reaction processing with parameter and structural priors (60), equilibrium with entity, parameter and structural priors (15 each), crystallization with entity priors (30) and purification with entity priors (15). Every block uses five worlds and three information arms. Electrochemistry and reaction processing have discovery and optimization commissions. Electrochemistry, partitioning and crystallization have 12- and 24-batch conditions; other blocks use 12. Parameter-prior P and purification P denote different concepts.

Electrochemistry varies materials, potential, current limits and duration; its optimization objective is a public balanced-efficiency score. Reaction processing varies materials, catalyst loading and thermal histories. Partitioning studies allocation under material, volume and process changes. Equilibrium studies amount, volume, concentration, staged additions, dissociation and precipitation. Crystallization seeks recovery net of seed mass subject to purity at least 0.80, fines fraction at most 0.50 and particles present. Purification seeks original-charge recovery subject to purity at least 0.80. Recovery denominators differ and are not combined.

Agent and information conditions

All source campaigns use GPT-5.6 Sol with medium reasoning effort in independent sessions. Within a block, arms share the physical world, operations, resource contract and withheld questions. The treatment modifies the public dossier according to the block's fixed entity, parameter or structural design. Opaque still exposes the operational interface and general task background. Aligned is a partial nominal description, such as a true effective-pKa interval in the equilibrium parameter study. MisIndexed applies the specified mismatch rather than supplying arbitrary misinformation.

Agents may choose conditions, repetitions, measurement allocations and experiment grouping, and may use the bounded public numerical tool. They are not required to choose from mechanism templates, emit a fixed equation or reflect after each batch. Batches, operations, provider responses and decisions made after feedback are different counts. Each budget permits a matching number of additional measurements and a final assay per completed batch; other resources remain separately bounded. The larger budget includes a larger execution and computation envelope. Retained early electrochemistry and partitioning assessments have an eight-call calculator allowance; later blocks disclose up to 128 attempts. Outputs are requested in

English, but retained reaction and equilibrium assessment prompts include Chinese instructions requesting English responses. These differences remain part of the historical protocol.

Sealed assessments and retests

Where required, a completed-batch recommendation is sealed before assessment and its procedure retested independently with a different observation seed in the same world. There are 165 such retests. K1 requests an unrestricted scientific account including relationships, evidence, applicability and unresolved alternatives. Q then requests point predictions and intervals for twelve fixed new-batch recipes without laboratory access or target feedback. K2 follows the sealed predictions with retrospective questions about support, limitations and discriminating experiments. The short version has three questions; reaction and equilibrium use seven themes. The bounded equilibrium parameter block adds fifteen structured supplements.

The original research context is available during every stage. Q can use the source history in addition to K1, and K2 cannot revise Q. Consequently, neither report-only knowledge transfer nor a uniform mechanism-correctness score is an endpoint. Selected textual examples illustrate observed behaviour; cohort-wide mechanism and trajectory annotation has not been completed.

Outcome definitions and references

For each response, campaign MAE averages absolute point error across the twelve questions. Interval coverage measures the fraction of specified reference targets contained in the reported intervals; width is their average span. Partitioning uses nominal 90% intervals and the other blocks use 80%. Partitioning and crystallization point references use underlying pre-final-assay response values. Electrochemistry and purification use fixed seeded observations. Reaction and equilibrium point references average five observations; their coverage is evaluated against the individual observations. Macro MAE averages response-specific MAEs equally within the designated block. These targets and scales are not pooled across systems.

The crystallization mean baseline uses only the source campaign’s public final assays. Nearest-neighbour prediction additionally uses public recipe features. A parser correction excludes resource-denied attempts from physical recipe features while preserving those attempts in the record. It restores two baseline evaluations; the other twenty-eight remain exactly unchanged. No agent or simulator is rerun for the correction. The equilibrium source-mean diagnostic is a separate post hoc reference. Its failure on dilute queries confirms that copying the plateau is insufficient for those questions; it is not a strong system-identification comparator.

Descriptive comparisons and retained failures

Goal comparisons match world, information arm, budget and prior locus. Budget comparisons match world, arm and goal, comparing independent sessions. Arm comparisons match the other factors within each study. Means weight campaigns equally. Five world clusters recur across arms, goals and budgets; questions, response metrics and batches are correlated observations. We report magnitudes, paired directions, all-arm tables and selected sensitivity analyses rather than inferential tests treating queries as independent trials.

The equilibrium dilution grouping was selected post hoc using the public concentrations of the three most dilute recipes, approximately 1.33×10^{-4} , 1.33×10^{-5} and 1.67×10^{-4} M. Grouped scores reproduce each original twelve-question macro score when weighted 3:9. The joint recovery–purity crystallization comparison is also exploratory. Neither changes the original query bank or replaces an unfavourable outcome. Retrospectively selected trace examples are distinguished from cohort-level findings.

All 240 campaigns remain included. One crystallization source has eleven rather than twelve final assays after solvent exhaustion. One purification source consumes twelve starts but has ten final assays and two discarded vessels. Both are marked nonconforming; neither is replaced. Infrastructure retries retain previous

attempts and resource accounting. The supplementary tables report the complete matrix, targets, qualification domains and every response-specific arm mean. Original protocol variation and excluded historical pilots are documented separately from this pool.

Data and code availability

The source observations, sealed outputs, evaluator-derived summaries and analysis code underlying this draft are retained in the project evidence records. The supplementary material reports aggregate outcomes and protocol details. A public archival package for this integrated study, including the applicable execution version and privacy-reviewed records, has not yet been finalized; the earlier platform release should not be assumed to contain these later studies.

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Supplementary information

A Protocol detail and evidence handling

A.1 Complete study matrix

System / prior locus	Scientific commission	Budgets	Campaigns	Final-assayed batches
EC / E: electrochemistry	Discovery; score optimization	12, 24	60	1,080 / 1,080
PA / E: phase partitioning	Explain phase allocation	12, 24	30	540 / 540
RX / P and S: reaction and thermal processing	Discovery; safety-constrained optimization	12	60	720 / 720
EQ / E: medium identity	Characterize entity-conditioned responses	12	15	180 / 180
EQ / P: bounded equilibrium	Characterize effective responses	12	15	180 / 180
EQ / S: structural equilibrium	Explain response structure	12	15	180 / 180
C / E: crystallization	Deliver crystals under quality constraints	12, 24	30	539 / 540
P / E: purification	Recover product subject to purity	12	15	178 / 180
Total	Six system families		240	3,597 / 3,600

Table A1. Complete selected study scope. Each campaign has one agent realization. One C campaign exhausts its solvent stock and discards its twelfth vessel, completing eleven final assays. One P campaign uses all twelve vessel starts but discards two vessels, leaving ten final assays. All campaigns have sealed mechanism, prediction, and reflection stages; 238 meet the full source-and-assessment specification.

A.2 Prediction targets and system-specific questions

System	Prediction responses	Reference target	Interval level
EC	Six conversion/efficiency responses, including score	Seeded final observation	80%
PA	Organic and aqueous product fractions	Noiseless pre-sampling fractions	90%
RX	Yield, conversion, selectivity, byproduct signal, risk, score	Five-observation mean; coverage over observations	80%
EQ / E, P and S	Normalized pH, dissociation, precipitation signal	Five-observation mean; coverage over observations	80%
C	Net recovery, purity, size index, fines fraction	Noiseless pre-assay responses	80%
P	Purity, original-charge recovery	Seeded final observation	80%

Table A2. Every campaign predicts twelve conditions. PA fractions are complementary, and its two decision questions reuse the conditions. Reference repeats reduce observation noise but do not add agent sessions.

EC’s twelve conditions balance positive and negative potential while varying duration, materials, and current cap at a stated loading. Its six responses are selective product yield, electrochemical selectivity, Faradaic efficiency, transport efficiency, energy efficiency, and public score. Current is a magnitude cap, not guaranteed delivered current. PA predicts phase fractions before analytical sampling; sampling losses and complementary fractions must not be counted as separate discoveries.

RX’s twelve paths examine time, temperature, catalyst, heating sequence, and quenching. Actual thermal telemetry matters: requested boundary temperature need not be realized reaction temperature. Five reference observations per condition are withheld from agents, with all source assessments sealed before reference-truth release. RX’s 600 reference executions are separate from 720 source batches and 60 retests.

Each EQ block uses twelve questions and five observations per question, giving 300 reference executions. EQ/E crosses three medium identities with three concentrations and a matched-concentration scale control, assessing entity contrast and concentration and scale transfer under a fixed common topology. Bounded EQ/P examines effective aqueous relationships; canonical EQ/S supports response-shape analysis without requiring a named hidden family. An instrument’s equilibrium diagnostic is not the agent’s subjective confidence. The bounded study’s effective-parameter supplement is additional to K1/Q/K2 and does not confer discovery credit for a parameter already supplied as a treatment.

C’s twelve conditions form six pairs: material, seed dose, cooling history, prior thermal history, continued growth, and upstream loading. Its size response is $\min(d_{50}/250\text{ }\mu\text{m}, 1)$, a bounded number-weighted index rather than a diameter in micrometres. Fines are particles below 20 micrometres. Quality requires purity and fines criteria jointly, with particles present; recovery of 0.10 is an initial feasibility target, not an optimization ceiling. P’s six paired interventions concern multistage recovery, including wash staging. Its original-charge denominator differs from C’s crystallization-stage recovery.

RX and EQ coverage uses individual reference observations while point targets use means. EC and P use single seeded observations. PA and C prediction coverage concerns noiseless pre-sampling responses; C recommendation retests use observed assays. Apparent retest threshold crossings can therefore reflect noise. Macro errors weight a system’s responses equally; normalized scales do not make responses physically or scientifically interchangeable.

A.3 Language, numerical tools, and stage budgets

Source briefs and requested reports are English. EC/PA and C/P assessment instructions are English; RX and EQ K1/Q/K2 instructions include Chinese text requesting English reports. Within-block arms have the same language contract. Seven-theme K2 reviews cover treatment claims, influential experiments, competing explanations, one proposed discriminating experiment, goal-related tradeoffs, unused evidence and uncertainty, and limitations of the recommendation or scientific account. Three-question versions combine these themes. The reviews occur after source experiments and sealed predictions; their length does not provide additional source experiments.

The earlier EC/PA block retains its saved eight-call diagnostic-calculator hard stop; later blocks disclose 128-attempt assessment allowances. An old EC parameter/structural supplement encountered the smaller limit and remains an execution failure outside the primary E cohort. The repair is not retroactively attributed to older sessions. Public calculation uses agent-accessible information, not hidden targets. Differences in numerical budgets and deadlines are execution contracts, not scientific findings.

A.4 Completion, recovery, and exclusions

The primary denominator is 240 scheduled campaigns, not provider attempts. All 720 K1/Q/K2 stages and 165 planned recommendation retests in EC, RX, C, and P are complete, plus fifteen EQ supplements. Final-assayed source batches total 3,597 of 3,600 planned. C’s second-world 12-batch Opaque source has eleven final assays. P’s fifth-world Opaque source starts twelve vessels, discards two, and completes ten final assays. Both remain in outcome tables with conformance indicators.

EC/PA has 68 first-attempt and 22 infrastructure-recovered campaigns. Recovery after transport or host failures retains earlier attempts and consumed resources. Missing assessment stages resume from saved context when the source is intact; fresh source replacements are recorded as such. Repeated question delivery and failed-attempt computation are not silently counted as an original clean attempt. EC goal sensitivity is reported in the main text. First-attempt-only EC-discovery, EC-optimization, and PA budget comparisons improve prediction in four of five, eight of eleven, and eight of ten pairs, respectively; these reduced subsets have uneven coverage.

The current P matrix follows a correction of prepartition inventory handling. Affected qualification and the entire agent block were restarted under the corrected contract; earlier apparently successful cells were not selected into the new matrix. All current source and recommendation trajectories pass inventory, resource, and replay checks. Its two lawful discards are a source-assay shortfall, not an inventory failure. The fixed purity-direction rule treats changes of 0.02 or less as small; a numerical tolerance of 10^{-12} corrects floating-point equality handling. Six direction classifications changed after this correction, with originals preserved. Predictions, point errors, intervals, retests, and denominators were unchanged.

Historical single-world EC parameter/structural supplements, the closed-set EQ/S pilot, early C/P development runs, qualification recipes, references, replays, and interrupted attempts do not enlarge the denominator. Their records remain separate. Frozen platform qualification is also a separate evidence programme. These counts describe execution and analysis scope; they are not 240 independent chemistry replications.

A.5 Baseline correction and trace interpretation

The C empirical references use the same source final observations for all three arms. A recipe parser originally rejected resource-denial transaction records in two sources. The correction preserves those attempts as provenance and excludes them from committed physical recipe features. Recomputing all thirty baselines restores the two missing evaluations and exactly reproduces the twenty-eight previously available results. Original trajectories, truth, model answers, and error records are retained; the correction uses no new simulator or provider calls. Unknown transaction statuses remain errors. This is an evaluator repair, not a change to a completed experiment.

The EC illustrative trace is the second-world, 12-batch Opaque EC discovery campaign; its sparse-positive-coverage counterexamples are first-world, 24-batch Aligned and MisIndexed discovery campaigns. They are retrospectively selected examples, not a coded prevalence estimate. The additional C case is the fifth-world, 12-batch MisIndexed campaign, chosen as the largest purity MAE in the 12-batch cohort. Its extreme error illustrates a failure; it is not a representative sampling rule.

A systematic analysis should attach each mechanistic claim to its experimental support, contradictory observations, stated domain, and corresponding predictions. It should distinguish an unperformed discriminating experiment, acquired evidence omitted from an account, justified unresolved alternatives, and report-prediction inconsistency. Evaluator knowledge helps identify discriminating interventions; a claim is not wrong merely for using an equivalent representation. Independent review can assist this analysis without replacing measured predictions and retests. No completed cohort-wide mechanism score or causal compression experiment is implied.

A.6 Full prior overview

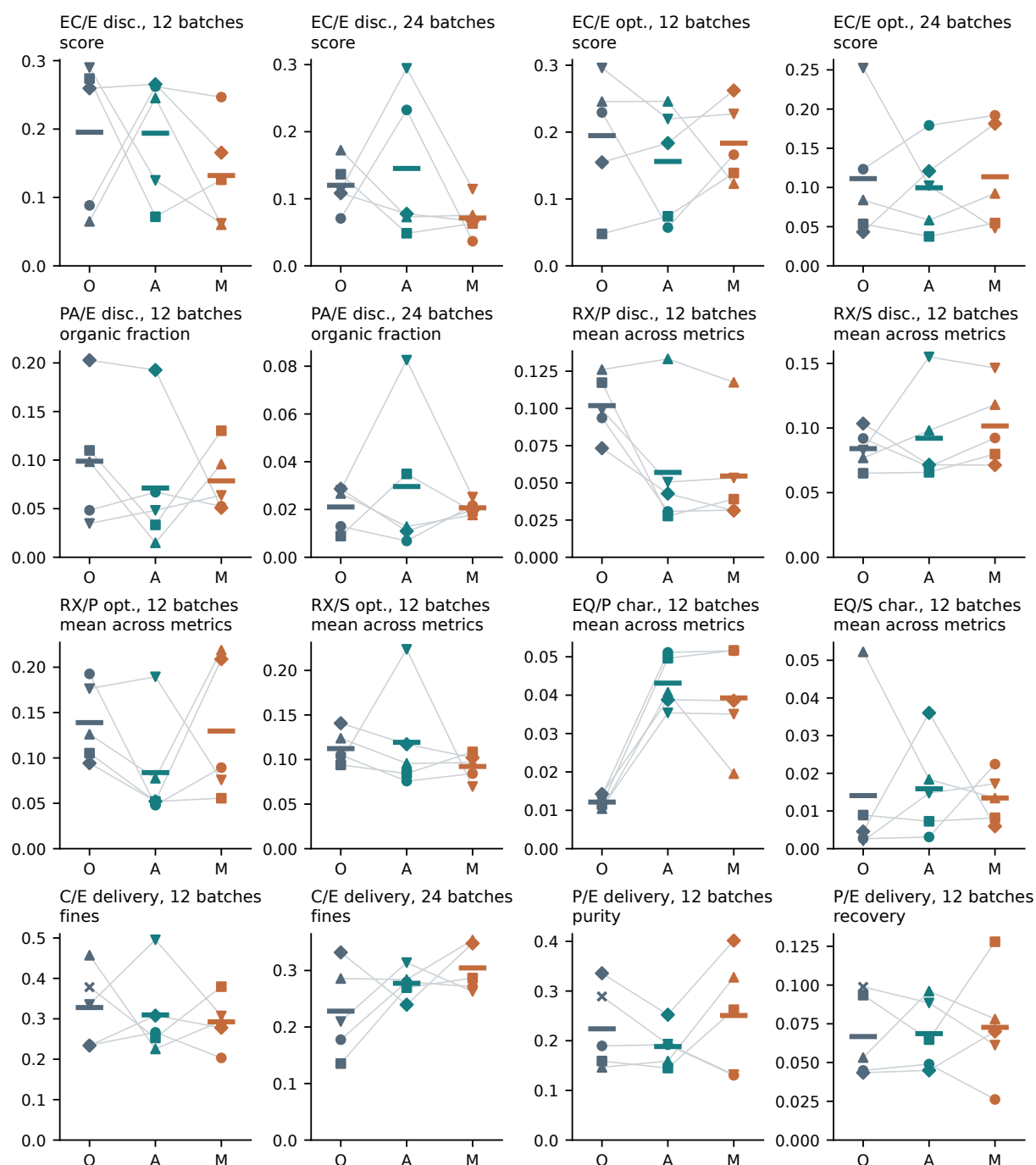


Figure 6: Prior information has no common ordering. Every panel includes five worlds and three arms: Opaque (O), Aligned (A), and MisIndexed (M). Gray lines connect worlds; colored bars show means. Crosses mark source shortfalls. RX and EQ use within-system macro errors; other panels name their response. Scales differ. Appendix B complements these selected readouts with all metric-specific arm means, including EQ/E.

B Complete metric-specific arm means

Each entry is mean absolute error / empirical interval coverage (%), averaged over five world sessions. All scheduled sessions are retained, including the two source-budget shortfalls. PA uses nominal 90% intervals; all other studies use 80%. Metric scales and reference targets differ across systems. The machine-readable table includes each campaign’s interval width and protocol status.

B.1 EC, E prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2983 / 55.0	0.2740 / 65.0	0.1272 / 88.3
Energy efficiency	0.3098 / 33.3	0.2595 / 51.7	0.1582 / 63.3
Faradaic efficiency	0.2580 / 46.7	0.2259 / 60.0	0.1356 / 73.3
Public score	0.1953 / 51.7	0.1939 / 60.0	0.1321 / 71.7
Selective product yield	0.0731 / 60.0	0.0817 / 38.3	0.0598 / 56.7
Transport efficiency	0.2602 / 45.0	0.2184 / 66.7	0.1328 / 76.7

B.2 EC, E prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.2908 / 63.3	0.2378 / 66.7	0.2661 / 73.3
Energy efficiency	0.2794 / 43.3	0.2541 / 31.7	0.2440 / 55.0
Faradaic efficiency	0.2506 / 53.3	0.2532 / 31.7	0.2315 / 56.7
Public score	0.1947 / 55.0	0.1561 / 58.3	0.1835 / 53.3
Selective product yield	0.0758 / 43.3	0.0623 / 53.3	0.0762 / 50.0
Transport efficiency	0.2478 / 58.3	0.2053 / 51.7	0.2217 / 60.0

B.3 PA, E prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3
Organic fraction	0.0989 / 76.7	0.0712 / 80.0	0.0786 / 78.3

B.4 EC, E prior: discovery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1902 / 68.3	0.1268 / 80.0	0.0428 / 93.3
Energy efficiency	0.1581 / 61.7	0.1651 / 68.3	0.0945 / 80.0
Faradaic efficiency	0.1458 / 65.0	0.1505 / 66.7	0.0877 / 78.3
Public score	0.1201 / 63.3	0.1452 / 73.3	0.0713 / 78.3
Selective product yield	0.0726 / 53.3	0.0573 / 70.0	0.0374 / 75.0
Transport efficiency	0.1495 / 65.0	0.1501 / 66.7	0.0846 / 78.3

B.5 EC, E prior: optimization, 24 batches

Readout	Opaque	Aligned	MisIndexed
Electrochemical selectivity	0.1403 / 85.0	0.1252 / 85.0	0.0864 / 90.0
Energy efficiency	0.1977 / 48.3	0.1307 / 60.0	0.1779 / 65.0
Faradaic efficiency	0.1502 / 76.7	0.1098 / 78.3	0.1515 / 71.7
Public score	0.1113 / 75.0	0.0996 / 71.7	0.1136 / 75.0
Selective product yield	0.0515 / 75.0	0.0498 / 56.7	0.0587 / 58.3
Transport efficiency	0.1420 / 80.0	0.1257 / 73.3	0.1570 / 68.3

B.6 PA, E prior: discovery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Aqueous fraction	0.0251 / 100.0	0.0296 / 96.7	0.0207 / 100.0
Organic fraction	0.0211 / 100.0	0.0296 / 96.7	0.0207 / 100.0

B.7 RX, P prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.0800 / 88.7	0.0791 / 73.3	0.0710 / 81.0
Conversion	0.0426 / 72.7	0.0321 / 70.0	0.0306 / 73.3
Safety risk	0.1031 / 58.3	0.0282 / 85.0	0.0370 / 81.7
Public score	0.1195 / 48.7	0.0501 / 83.3	0.0476 / 90.3
Selectivity	0.1198 / 58.0	0.0818 / 69.3	0.0714 / 77.3
Yield	0.1461 / 35.7	0.0708 / 76.7	0.0698 / 78.3

B.8 RX, P prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1509 / 57.7	0.0805 / 73.0	0.1271 / 60.3
Conversion	0.0474 / 85.3	0.0319 / 82.7	0.1960 / 53.0
Safety risk	0.0984 / 56.7	0.1106 / 56.7	0.0811 / 68.3
Public score	0.1341 / 30.0	0.0818 / 73.0	0.0879 / 68.3
Selectivity	0.1900 / 31.0	0.0923 / 69.7	0.1360 / 64.7
Yield	0.2128 / 19.3	0.1060 / 65.7	0.1499 / 68.7

B.9 RX, S prior: discovery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1219 / 78.7	0.1212 / 75.3	0.1279 / 64.7
Conversion	0.0217 / 97.7	0.0266 / 95.3	0.0200 / 98.0
Safety risk	0.0664 / 80.0	0.0717 / 66.7	0.1276 / 43.3
Public score	0.0504 / 89.7	0.0816 / 61.3	0.0665 / 78.0
Selectivity	0.1280 / 72.7	0.1319 / 70.7	0.1400 / 55.0
Yield	0.1157 / 81.7	0.1196 / 75.7	0.1275 / 61.0

B.10 RX, S prior: optimization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Byproduct signal	0.1376 / 63.7	0.1490 / 67.7	0.1188 / 66.3
Conversion	0.0370 / 88.7	0.0505 / 82.7	0.0425 / 88.3
Safety risk	0.1000 / 45.0	0.1255 / 68.3	0.1133 / 45.0
Public score	0.0932 / 56.0	0.0917 / 58.0	0.0637 / 81.0
Selectivity	0.1574 / 56.7	0.1570 / 60.3	0.1175 / 62.3
Yield	0.1485 / 63.3	0.1422 / 66.3	0.0975 / 73.0

B.11 EQ, P prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0129 / 96.3	0.0759 / 68.7	0.0646 / 74.7
Normalized pH	0.0031 / 94.3	0.0160 / 65.0	0.0150 / 68.7
Precipitation signal	0.0205 / 83.3	0.0376 / 73.3	0.0383 / 68.7

B.12 EQ, S prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0292 / 91.7	0.0235 / 76.3	0.0122 / 87.7
Normalized pH	0.0097 / 87.3	0.0200 / 75.7	0.0203 / 77.7
Precipitation signal	0.0033 / 95.3	0.0042 / 95.7	0.0078 / 87.3

B.13 EQ, E prior: characterization, 12 batches

Readout	Opaque	Aligned	MisIndexed
Dissociation fraction	0.0128 / 83.7	0.0107 / 85.0	0.0134 / 79.0
Normalized pH	0.0233 / 80.7	0.0278 / 79.0	0.0202 / 82.0
Precipitation signal	0.0084 / 89.3	0.0116 / 92.0	0.0103 / 88.3

B.14 C, E prior: delivery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.3279 / 25.0	0.3096 / 38.3	0.2925 / 43.3
Crystal purity	0.0095 / 98.3	0.0497 / 60.0	0.0917 / 23.3
Particle-size index	0.0777 / 55.0	0.0624 / 58.3	0.0725 / 58.3
Net crystal recovery	0.1632 / 48.3	0.0606 / 83.3	0.0892 / 71.7

B.15 C, E prior: delivery, 24 batches

Readout	Opaque	Aligned	MisIndexed
Fines fraction	0.2281 / 51.7	0.2774 / 38.3	0.3046 / 41.7
Crystal purity	0.0197 / 95.0	0.0398 / 73.3	0.0508 / 55.0
Particle-size index	0.0566 / 71.7	0.0440 / 70.0	0.1023 / 48.3
Net crystal recovery	0.0911 / 61.7	0.0684 / 80.0	0.0786 / 73.3

B.16 P, E prior: delivery, 12 batches

Readout	Opaque	Aligned	MisIndexed
Purity	0.2240 / 51.7	0.1883 / 53.3	0.2508 / 50.0
Recovery	0.0668 / 56.7	0.0687 / 51.7	0.0727 / 53.3

C Exploratory analyses of response regimes

These analyses use the retained source campaigns and their original predictions. They do not define new experimental arms or alter the withheld-condition bank. The concentration grouping, joint crystallization contrast and illustrative traces were selected after the experiments, and are descriptive analyses of reused worlds.

C.1 Equilibrium: a reversal across public concentration regimes

The three most dilute recipes have nominal concentrations approximately 1.33×10^{-4} , 1.33×10^{-5} and 1.67×10^{-4} M. Each campaign has three normalized responses for each recipe. Point errors use five-observation means; interval coverage uses the individual reference observations. Group means average the corresponding responses and queries and then the five world campaigns. Weighting the two groups 3:9 reconstructs every original twelve-question campaign macro MAE. The other nine queries include boundary conditions; this split is not a predeclared interpolation-versus-extrapolation test.

Query group / arm	Macro MAE	Coverage (%)	Interval width	Source-mean MAE
Other nine / Opaque	0.01008	90.7	0.04706	0.00553
Other nine / Aligned	0.00465	91.3	0.03340	0.00518
Other nine / MisIndexed	0.00439	89.6	0.02798	0.00494
Three most dilute / Opaque	0.01833	93.3	0.15676	0.16961
Three most dilute / Aligned	0.15862	2.2	0.08084	0.16833
Three most dilute / MisIndexed	0.14397	13.8	0.09633	0.16836

Table C1. EQ/P subgroup results. Each row averages five worlds; rows reuse those worlds. The source-mean predictor is an additional post hoc diagnostic and obtains no new experiments. Its poor dilute-query performance shows why retaining the plateau is insufficient; its usefulness on the other nine depends on their response distribution.

Aligned improves on the other nine and worsens on the dilute three relative to Opaque in all five worlds. All fifteen source campaigns remain above 0.018 M. In the selected second-world comparison, Aligned spans about ninety-fold source concentration variation and Opaque about fourfold, but both remain in the observed plateau. At the most dilute question, reference dissociation is 0.65463; Opaque predicts 0.629 [0.45, 0.80] and Aligned 0.0804 [0.053, 0.108]. The Opaque prediction response develops an unsaturated weak-acid extrapolation beyond its sealed report. This reasoning must not be credited retrospectively to K1 or to an earlier source-stage hypothesis.

C.2 Crystallization: joint response changes under the same treatment

Budget / arm	Recovery MAE	Purity MAE	Recovery improves and purity worsens vs Opaque
12 / Opaque	0.16322	0.00952	Reference
12 / Aligned	0.06055	0.04969	5 / 5 worlds
12 / MisIndexed	0.08923	0.09166	5 / 5 worlds
24 / Opaque	0.09107	0.01975	Reference
24 / Aligned	0.06839	0.03983	2 / 5 worlds
24 / MisIndexed	0.07863	0.05080	3 / 5 worlds

Table C2. Joint directional comparisons use both response errors from the same world, arm and budget. At twelve batches, each joint direction also holds in all four pairs whose two campaigns conform to the source-assay contract. The direction weakens at 24 batches. A treatment can therefore be helpful on one response and harmful on another without implying a budget-invariant tradeoff.

C.3 Evidence status and interpretation

The same-source C references use only each campaign’s final assays and public recipe features. They exclude hidden truth during fitting but do not reproduce the agent’s complete intermediate measurements, prior or computational policy. Low held-out purity variation makes the mean a particularly strong reference. Neither its success nor the agent’s recovery advantage is a complete mechanism identification result.

Selected EC, C and EQ traces support detailed examples rather than population frequencies. The current evidence does not causally identify prior anchoring, report compression, acquisition-versus-inference mediation or a general preference for familiar chemistry. Full-history Q also prevents interpreting K1 as the sole information channel. A cohort-wide, claim-level annotation and prospective evidence-matched interventions remain future analyses.